

Satellite-Based Urban Growth and Economic Activity Intelligence System

Review 2 — Project Documentation

BCSE497J Project I

School of Computer Science and Engineering

Fall Semester 2026–27

Study area — Varanasi, Uttar Pradesh, India

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Guide — [Guide Name]

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Abbreviations and official dataset identifiers are listed in `CONVENTIONS.md`. Every figure in this document comes from an actual pipeline run recorded in `outputs/varanasi_summary.json`.

1. Abstract

Indian cities are expanding faster than their populations, and municipal authorities have no systematic way to tell development that is being occupied from development that is not. In Varanasi, built-up surface grew **23.8%** between 2010 and 2020 while population grew **10.3%** — land consumed roughly **2.3 times faster** than people were housed. Expansion is normally measured by area alone, and area cannot distinguish a functioning new neighbourhood from an empty one. Municipalities therefore extend water, sewerage and transport on the basis of construction rather than occupancy, with no instrument for detecting when that investment is stranded.

This project builds an end-to-end system that measures urban expansion, commercial activity, green-cover loss and surface urban heat from open satellite and geospatial data, and combines them to identify **ghost growth** — land that has been built on but shows economic activity far below the level normal for comparably developed land in the same city.

Built-up surface is taken from [GHS-BUILT-S R2023A](#) across the observational epochs 2010, 2015 and 2020, and every newly urbanised cell is classified as infill, edge expansion or leapfrog using a landscape-expansion-index method. Economic activity is estimated from three independent signals — VIIRS nighttime radiance, OpenStreetMap point-of-interest density, and gridded population — each normalised per unit of built-up area. Rather than applying a fixed radiance threshold, the system **learns the activity level expected at each built-up intensity for that specific city** and flags the negative residuals, which makes the method transferable to any city without re-tuning. A logistic suitability model coupled to a constrained cellular automaton then projects where expansion will occur next, trained on one period and validated on a later period it has never seen.

The distinction that matters most is between development that has stalled and development still filling up. Without a nighttime-light time series the system flagged **7.24 km²** as underused; adding the 2013–2024 series split that almost exactly into **0.35 km² genuinely stalled** and **6.88 km² still being occupied**. A method that cannot draw this distinction overstates the problem roughly twentyfold.

The system runs end-to-end on data requiring no credentials, and folds in Google Earth Engine layers automatically once authentication is available. Results are delivered through an interactive dashboard and a machine-readable grid export carrying full provenance.

Keywords — remote sensing, urban growth, nighttime lights, machine learning, cellular automata, Google Earth Engine, land-change modelling.

2. Literature Review

Twenty papers were reviewed, **nineteen of them published in 2020 or later**. Each had to implement a model, release or report a product, and give quantitative validation; reviews and papers that merely consume such products were excluded. Per-paper summaries are in `PAPER_SUMMARIES.md` and the full thematic treatment is in `LITERATURE_REVIEW.md`.

2.1 Theme A — Nighttime light as a proxy for human activity

Nighttime light radiance is the most widely used indirect indicator of human and economic activity. The difficulty is that the long historical record comes from two incompatible sensors: DMSP-OLS (Defense Meteorological Satellite Program — Operational Linescan System), covering 1992–2013 but saturating in bright cores and spreading light beyond its true source, and VIIRS (Visible Infrared Imaging Radiometer Suite), covering 2012 onward with better calibration and a far wider measurable range.

Study	Year	Method	Reported result
Li, X. et al.	2020	Sigmoid function	The standard harmonised global series, 1992–2018
Chen, Z. et al.	2021	Auto-encoder network plus a vegetation index	R^2 0.87 per pixel, 0.95 per city
Nechaev, D. et al.	2021	Residual U-Net	R^2 0.94–0.99
Chen, X. et al.	2024	U-Net as image super-resolution	R^2 0.617 per pixel rising to 0.964 per country
Zhang, L. et al.	2024	Convolutional Long Short-Term Memory network	RMSE 0.73, R^2 0.95 per pixel
Tian, Y. et al.	2026	Two-stage deep model guided by impervious surface	R^2 0.8088, beating both earlier products

Finding 1 — accuracy depends on scale, not on the model. Chen, X. et al. report R^2 rising 0.617 → 0.747 → 0.874 → 0.964 from pixel to city to province to country. Chen, Z. et al. show the same pattern. Six studies span sigmoid functions, auto-encoders, U-Nets, Long Short-Term Memory networks and two-stage deep models, and within a fixed transform direction they land in the same accuracy band. Chen, X. et al. describe their own network as performing "closely to" an auto-encoder.

Finding 2 — the difficulty lies in the direction, not the architecture. Converting VIIRS down to look like DMSP-OLS is a monotone mapping plus a blur, so a sigmoid suffices. Rebuilding VIIRS-like detail *from* DMSP-OLS is under-determined, because the detail was never recorded. Both studies that succeed at the hard direction add a **daytime** channel — a vegetation index in Chen, Z. et al., Landsat NDVI in Chen, X. et al., impervious surface in Tian et al. The auxiliary data, not the network, is the enabling ingredient.

Finding 3 — residual error concentrates at the urban fringe. Chen, X. et al. document that their product underestimates radiance at the city edge, because the network pulls bright pixels toward darker neighbours, while the competing ChenVNL product overestimates cores and underestimates peri-urban land. Two networks, two bias signs, both worst at the fringe.

2.2 Theme B — Measuring built-up extent

Study	Year	Data	Result
Goldblatt, R. et al.	2018	Landsat + nighttime light	80.8% balanced accuracy for India
Marconcini, M. et al.	2020	Landsat + Sentinel-1 radar	Kappa 0.6885; +0.23 over GHSL
Tang, Y. et al.	2021	Nighttime light + MODIS	R ² 0.9239 for sealed-surface percentage
Sirko, W. et al.	2021	U-Net on 50 cm imagery	516 million building footprints
Brown, C. F. et al.	2022	Deep learning on Sentinel-2	First near-real-time global land cover

Goldblatt et al. is the methodological ancestor of this project: it uses nighttime light to label training data automatically, avoiding hand-drawn training areas, and was validated on India. Sirko et al. found that **deeper encoders gave no accuracy gain** — improvements came from training technique rather than architecture. Marconcini et al. quantify where GHSL is weakest, namely small and scattered settlements, which is a documented limitation of the dataset this project relies on.

What this theme leaves open. All five produce a map of *where buildings are*. None says anything about whether those buildings are used.

2.3 Theme C — Inferring activity and vacancy

Zhang, Y., Tu, T. & Long, Y. (2024) is the closest published work. Across 8,841 cities worldwide they split each into areas developed before and after 2005 and measured urban vitality from road density, point-of-interest density and population density. New areas showed only **7.69%** of the vitality of older areas, and the worst-scoring 5% — 442 cities — were labelled ghost cities.

Yeh, C. et al. (2020) trained deep networks on public satellite imagery across about 20,000 African villages and explained **70%** of the variation in ground-measured wealth in countries the model had never seen. Their targeting accuracy was 81% using multispectral imagery plus nighttime light against 62% using nighttime light alone — direct evidence for combining signals.

Anucharn, T. et al. (2025) provide a working Google Earth Engine template, reporting overall accuracy of 0.80–0.82 against 0.73–0.76 for traditional classification, and R² 0.9744 between nighttime light and electricity consumption. That correlation is between **provincial totals**, not pixels, and treating an aggregate correlation as evidence about individual pixels is a common error this project explicitly avoids.

2.4 Theme D — Predicting future urban expansion

Study	Year	Method	Result
Chen, G. et al.	2020	Scenario projection to 2100	50–63% of new urban land falls on cropland
Liang, X. et al.	2021	Random forest + patch-based cellular automaton	Figure of Merit 0.2642
Shojaei, H. et al.	2022	Modified U-Net	AUC 0.87 against random forest 0.82
Zhang, A. et al.	2025	Earth Engine + cellular automaton with neural network	Classification accuracy above 92%

Liang et al. established the pattern this project follows — learn where growth is likely, then use a cellular automaton so new development appears in connected patches. Their Figure of Merit of 0.2642 is the benchmark our own result is measured against. Chen, G. et al. chose the Figure of Merit explicitly because conventional measures such as the Kappa coefficient overstate accuracy in land change, which is the same argument this project makes.

Shojaei et al. is the closest analogue to our prediction module: almost the same driver variables (altitude, slope, distance to roads and urban areas) and the same decade-scale epoch spacing.

Finding 4 — deep learning buys automation, not accuracy. Wang et al. (2022) report U-Net at parity with cellular-automaton models. Shojaei et al. beat a random forest by 0.05 in area under the curve — and a random forest has no view of a cell's surroundings at all, so much of that gap reflects spatial context rather than deep learning as such. Sirko et al. found deeper networks gave nothing. What a U-Net buys is automation and transferability; what it costs is labelled data, a graphics processing unit, and interpretability.

2.5 Theme E — Environmental consequences

Ermida, S. L. et al. (2020) published open Google Earth Engine code for computing LST (Land Surface Temperature) from Landsat, validated against ground radiometers at RMSE 1.0–1.3 K. This is the reference method behind our thermal layer, and it tells us how large a heat-island difference must be before it is worth reporting.

Ramachandra, T. V. et al. (2025) map urban heat islands against landscape morphology for an Indian city, finding 15.41 km² at very high temperature and a relationship between land cover and temperature of R² 0.49 — a useful reminder that land cover explains only part of surface temperature.

2.6 Research gap

What exists	What it does	What it does not do
Nighttime-light series (A)	Long, consistent activity records	Say nothing about whether construction is occupied
Built-up mapping (B)	Show precisely where buildings are	Show whether those buildings are used
Ghost-city studies (C)	Compare new against old, or use a fixed cut-off	Adapt to each city's own norm; separate stalled from filling-up
Expansion models (D)	Predict <i>where</i> a city grows	Ask whether that growth will be used

The gap. No existing study learns the activity level normal for a given level of development *within a city* and uses it to separate development that has stalled from development that is still being occupied.

This is not a theoretical gap. Section 7.4 shows that on our own data the distinction is the difference between 7.24 km² and 0.35 km².

3. Methodology

3.1 Problem framing

The two obvious detectors both fail:

Detector	What it actually flags
Low nighttime light	Every agricultural field, every unlit industrial estate
Low population density	Every warehouse district, every commercial zone

Both mistake **different** for **empty**. A correct method must (i) learn what normal activity is *for that city* at each level of development, (ii) flag departures from that norm rather than from a global threshold, and (iii) separate stalled development from development mid-occupation.

3.2 Spatial frame

Two grids are constructed together from a single origin.

Grid	Cell size	Dimensions	Used for
Fine	100 m	365 × 345	All analysis
Coarse	500 m	73 × 69 → 4,765 cells with signal	All reporting and display

Exactly 25 fine cells nest inside one coarse cell. Analysis runs at 100 m because that is the native resolution of `GHS-BUILT-S R2023A` and change detection works best at the finest resolution available. Reporting runs at 500 m because a map of the study area at 100 m has more than 121,000 cells, which is unusable in a browser and would imply more precision than the sensors support — the nighttime-light sensor resolves about 460 m at best.

Both grids must be created together. If each were snapped independently to its own cell size, they would not tile one another and every aggregate would be silently shifted. This is implemented in `AOI.frame_pair()`.

3.3 Density-preserving reprojection

GHSL products ship in Mollweide (ESRI:54009), an equal-area projection; the analysis frame is UTM zone 44N (EPSG:32644). Built-up surface in m² and population counts are **extensive** quantities — they scale with cell area — so each is converted to a density, reprojected, and rescaled by the destination cell area. Reprojecting the raw counts directly would not conserve totals.

3.4 Built-up expansion and urban form

Source. `GHS-BUILT-S R2023A`, observational epochs 2010, 2015 and 2020.

Change is computed by differencing epochs. Each cell that newly crosses the 20% built-up threshold is then classified by its spatial relationship to the existing urban fabric, following the landscape expansion index of

Liu et al. (2010):

Class	Definition	Why it matters
Infill	New development inside the existing urban area	Cheapest to service
Edge expansion	New development touching the existing edge	Moderate cost
Leapfrog	New development detached from existing fabric	Most expensive to service, most associated with under-occupancy

Leapfrog share is the leading indicator for ghost growth: detached development is exactly where built-but-inactive land should concentrate.

3.5 The activity index

Three independent signals are combined, each normalised **per unit of built-up area** so the value reflects intensity of use rather than size of settlement.

Signal	Weight	Source
Nighttime light radiance	0.45	NOAA/VIIRS/DNB/ANNUAL_V21 and _V22
Point-of-interest density	0.35	OpenStreetMap via the Overpass API
Population density	0.20	GHS-POP R2023A

Normalisation is essential: raw radiance mostly measures how large a place is. Three signals are used rather than one because each fails differently — light misses unlit but occupied areas, points of interest are unevenly mapped, and population misses commercial districts. Requiring agreement across signals is what makes a low reading meaningful.

3.6 The ghost-growth method — the project's differentiator

Step 1 — learn the city's own norm

Instead of a fixed threshold, the system computes a **binned median of activity against built-up intensity** across the whole city. This produces an *expected-activity curve* describing what activity level is normal for each level of development **in Varanasi specifically**.

This is the core methodological contribution. A fixed brightness cut-off has to be re-tuned for every city and every sensor version. A learned norm transfers to any city unchanged, because it is derived from that city's own data.

Step 2 — residual and recency screen

A cell is flagged when **both** hold:

- its activity falls far below the expected curve (a large negative residual), and
- the development is recent — measured by `new_share`, the proportion of the cell's current built-up area that appeared after the 2010 baseline.

The recency condition prevents long-standing low-activity land (an old industrial estate) from being confused with new development that never filled.

Step 3 — the trend test that separates stalled from filling-up

Where a nighttime-light time series exists, each cell's radiance trend is fitted over 2013–2024. Cells that are dim **but measurably brightening** are classified `emerging` rather than `ghost_growth`.

This single test is what §2.6 identifies as missing from the literature, and §7.4 shows it accounts for 95% of what would otherwise be reported as a problem.

Step 4 — the six-class growth typology

Class	Meaning
<code>ghost_growth</code>	New development, activity far below expectation, not rising
<code>emerging</code>	New development, low activity but rising — filling up
<code>healthy_growth</code>	New development performing normally
<code>established_active</code>	Built before the baseline, performing normally
<code>declining</code>	Established, underperforming and falling
<code>undeveloped</code>	Below the built-up threshold

Step 5 — zone clustering

Individual flagged cells are grouped into contiguous zones using a **density-ratio-to-background** rule: a neighbourhood qualifies when its local density of flagged cells exceeds the citywide background rate by a set factor. This produces zones a planner can inspect, rather than scattered single cells.

3.7 Green cover and surface heat

Green cover. NDVI (Normalized Difference Vegetation Index) is computed from `COPERNICUS/S2_SR_HARMONIZED` over an October–March window, avoiding the heavily clouded June–September monsoon. Only NDVI decline that **coincides with built-up gain** is counted as urbanisation-driven, because most vegetation change across this largely agricultural area is the *rabi* cropping calendar rather than construction.

Surface heat. LST is computed from `LANDSAT/LC08/C02/T1_L2` and `LANDSAT/LC09/C02/T1_L2` band `ST_B10` over the pre-monsoon March–May window. Heat-island intensity is LST minus the median LST of rural land **in the same scene, with water excluded**. Excluding water matters: the Ganga would otherwise drag the rural baseline down and inflate apparent heat-island intensity across the whole city.

3.8 The predictive expansion model

Drivers

Seven variables are computed per cell:

Driver	Rationale
Distance from the city centre	Growth concentrates nearer the centre
Distance from the existing built-up edge	New building appears beside old building
Current built-up fraction	Partly built cells fill in further
Road density	Roads precede buildings
Population	People attract construction
Neighbourhood built-up	Local context drives conversion
Slope	Steep land is harder to build on

Model

Logistic Regression estimates, for each currently non-urban cell, the probability that it becomes built-up. Class weights are balanced, because conversion is a rare event.

A **constrained cellular automaton** then allocates the expected quantity of new development to the highest-suitability cells, with each cell also weighted by its neighbourhood (300 m neighbourhood, neighbourhood weight 0.35, eight iterations). Without this step the model scatters predicted growth as isolated cells; with it, development appears in connected patches, which is how cities actually grow.

Why not deep learning

Per §2.4: published work reports U-Net at parity with cellular-automaton models for this task, the gain over a random forest is around 0.05 in area under the curve, and deeper networks gave no gain at all in Sirko et al. A U-Net needs a large labelled training set and a graphics processing unit, and produces a model whose reasoning cannot be shown to a planner. Logistic Regression needs neither and yields coefficients that can be read directly. Deep learning is recorded as a possible later upgrade, not dismissed.

Validation discipline

- **Train** on 2010→2015. **Test** on 2015→2020, a period the model never saw.
- **Benchmark** against random allocation of the same demand, repeated 20 times.
- **Epoch discipline.** [GHS-BUILT-S R2023A](#) supplies epochs to 2030, but only 1975–2020 are observational; 2025 and 2030 are the GHSL model's own projections. Fitting or validating against them would measure agreement between two models rather than accuracy against reality, so they are excluded from all fitting.

Why Figure of Merit rather than accuracy

$$\text{Figure of Merit} = \text{hits} / (\text{hits} + \text{misses} + \text{false alarms})$$

Overall accuracy is actively misleading for land change. On this data a **null model predicting no change at all scores 98.91%** — better than our model's 98.1%. Persistence dominates every cell count. The Figure of Merit ignores correct rejections and measures only the change class, which is what is being predicted. Chen, G. et al. (2020) give the same justification.

4. Dataset Exploration

4.1 Datasets requiring no account

Official product	Provider	Resolution	Coverage used
<code>GHS-BUILT-S R2023A</code>	European Commission Joint Research Centre	100 m	1975–2020 observed, five-yearly
<code>GHS-POP R2023A</code>	European Commission Joint Research Centre	100 m	1975–2020, five-yearly
OpenStreetMap via the Overpass API	OpenStreetMap contributors	vector	Current snapshot

`GHS-BUILT-S R2023A` is the backbone because it is the only multi-epoch, globally consistent built-up product downloadable in bulk **without an account**, and its five-yearly epochs land exactly on the years we want. It is distributed on a 1,000,000 m Mollweide grid; Varanasi falls entirely inside tile `R6_C26`, computed from the area of interest rather than hard-coded.

4.2 Datasets requiring Google Earth Engine

Official asset identifier	Provider	Resolution	Imagery dates
<code>NOAA/VIIRS/DNB/ANNUAL_V21</code>	NOAA	463 m	2013–2021
<code>NOAA/VIIRS/DNB/ANNUAL_V22</code>	NOAA	463 m	2022 onward
<code>COPERNICUS/S2_SR_HARMONIZED</code>	ESA / Copernicus	10 m native	01 Oct 2024 – 30 Mar 2025
<code>LANDSAT/LC08/C02/T1_L2</code>	USGS	30 m	12 Mar – 15 May 2024
<code>LANDSAT/LC09/C02/T1_L2</code>	USGS	30 m	2021 onward
<code>GOOGLE/DYNAMICWORLD/V1</code>	Google	10 m	01 Oct 2024 – 30 Mar 2025
<code>GOOGLE/Research/open-buildings-temporal/v1</code>	Google	4 m	2023
<code>USGS/SRTMGL1_003</code>	NASA / USGS	30 m	Single epoch

Acquisition dates were **read back from the image collections themselves**, not assumed from filenames. The Sentinel-2 composite is a median of **116 scenes across 29 separate days**; the Landsat thermal composite is **10 scenes on 5 days**; Dynamic World is **154 classifications on 60 days**.

4.3 Two VIIRS assets are required, not one

The annual VIIRS series is **split across two product versions** in the Earth Engine catalogue. `ANNUAL_V21` holds 2013–2021 and `ANNUAL_V22` holds 2022 onward. Querying `ANNUAL_V22` for an earlier year returns an **empty collection rather than an error**. The two versions share the same core compositing algorithm, so joining them is a version step rather than a sensor change — but it is still a discontinuity, and any trend crossing 2021/2022 carries it.

4.4 Bands actually used, and why

Dataset	Band	Reason
NOAA/VIIRS/DNB/ANNUAL_V21 / _V22	average_masked	The plain average band retains a background noise floor that registers as activity in unlit peri-urban cells — exactly the cells being judged
COPERNICUS/S2_SR_HARMONIZED	B8 , B4	Near Infrared and Red, giving NDVI
COPERNICUS/S2_SR_HARMONIZED	B11 , B8	Shortwave Infrared and Near Infrared, giving NDBI (Normalized Difference Built-up Index)
COPERNICUS/S2_SR_HARMONIZED	B4 , B3 , B2	Red, Green, Blue — true-colour composite
LANDSAT/LC08/C02/T1_L2	ST_B10	Surface temperature band of Collection 2 Level-2

4.5 Sentinel-2 coverage is sparse before 2018

October–March window	Scenes	Distinct days
2015–16	3	1
2017–18	57	18
2018–19	117	31
2024–25	148	35

A median composite over a single day is not a seasonal median — it is one observation with that day's weather and phenology baked in. The vegetation baseline is therefore **2018**, the first year past 30 acquisition days, and the code refuses any composite built from fewer than 20 distinct dates.

4.6 Dataset, raw variable, derived output

These three are distinct and are never used interchangeably.

Official dataset	Raw variable	Derived output
NOAA/VIIRS/DNB/ANNUAL_V21 / _V22	Nighttime light radiance	Activity map; activity trend
COPERNICUS/S2_SR_HARMONIZED	Surface reflectance	NDVI → vegetation map → green-cover change
LANDSAT/LC08/C02/T1_L2	Surface temperature	Heat-island intensity map
GHS-BUILT-S R2023A	Built-up surface density	Expansion map; infill/edge/leapfrog classification
GHS-POP R2023A	Gridded population estimate	Activity denominator
OpenStreetMap	Mapped points and lines	Point-of-interest density; road density

4.7 What is measured and what is inferred

No dataset used here contains economic output, occupancy or ownership. Every statement about those is an inference and is labelled as one.

Claim	Status
Built-up surface area	Measured , with model error
Nighttime light radiance	Measured
Land surface temperature	Measured
Economic activity	Proxy — radiance, points of interest and population combined
Ghost growth	Inference — activity far below the local norm
Population	Estimate — GHS-POP R2023A is modelled, not a census
Built-up in 2025 and 2030	Projection — GHSL's own model output
Urban extent in 2030	Prediction — our suitability model plus cellular automaton

4.8 Datasets deliberately not used

Dataset	Why not
NOAA/DMSP-OLS/NIGHTTIME_LIGHTS	The analysis window starts in 2010 and the baseline is 2010. DMSP-OLS saturates and blooms, and converting it to match VIIRS adds a large error term to buy history the project does not use
WorldPop	GHS-POP R2023A is built from the same chain as GHS-BUILT-S , keeping population and built-up internally consistent. WorldPop is implemented as an independent cross-check but is off the critical path
COPERNICUS/S1_GRD	Sentinel-1 radar needs a speckle-filtering and calibration chain the project does not require, given GHSL already supplies validated built-up surface
NASA/HLS/HLSL30/v002	COPERNICUS/S2_SR_HARMONIZED alone gives enough clear October–March scenes
GHS-SMOD R2023A	Itself derived from GHS-BUILT-S and GHS-POP by thresholding. The two masks needed are derived directly — same information, about 1 GB less transfer, and the masks stay at 100 m instead of a 1 km floor

5. System Architecture

5.1 Four layers



Data flows upward; each layer talks only to the one below it.

5.2 Module map

Module	Responsibility
<code>config.py</code>	Reads the configuration file
<code>aoi.py</code>	Builds the two matching grids from the area of interest
<code>data/ghsl.py</code>	<code>GHS-BUILT-S R2023A</code> and <code>GHS-POP R2023A</code> , including tile computation
<code>data/osm.py</code>	OpenStreetMap places and roads via the Overpass API
<code>data/gee.py</code>	All Google Earth Engine layers
<code>data/worldpop.py</code>	Optional independent population cross-check
<code>data/download.py</code>	Resumable downloads, extraction
<code>analysis/builtup.py</code>	Change detection, urban form, growth hotspots
<code>analysis/nightlights.py</code>	Trend fitting, Sum of Lights, normalisation
<code>analysis/vegetation.py</code>	Green-cover change and conversion to built-up
<code>analysis/thermal.py</code>	Heat-island intensity, vulnerability
<code>analysis/ghost.py</code>	Activity index, expected-activity residual, typology
<code>analysis/growth_model.py</code>	Driver construction, fitting, allocation, projection
<code>analysis/zonal.py</code>	100 m → 500 m aggregation and export
<code>pipeline.py</code>	Runs the stages in order
<code>dashboard/app.py</code>	The dashboard

Two rules keep the boundary clean:

- **Acquisition modules never analyse.** They fetch and reproject, nothing else.
- **Analysis modules never download.** They receive grids already prepared.

This is why analysis can be tested on synthetic data with no network access — the test suite runs in seconds.

5.3 Inter-stage data model

Each stage adds to one shared object passed along the pipeline:

Field	Contents
<code>layers</code>	The grids themselves, one per indicator
<code>stats</code>	Summary numbers for the report
<code>provenance</code>	Which dataset each layer came from
<code>skipped</code>	Any layer that could not be produced, and why

`skipped` is the important one. A layer that fails is **recorded rather than dropped**, so a missing result is visible in the output and on the dashboard instead of silently reading as zero.

5.4 Failure behaviour

stage runs

- └ success → results added, next stage starts
- └ failure → reason recorded in `skipped`, next stage starts

Public data sources rate-limit and time out, and Google Earth Engine requires an account. If any one of those halted the whole run, the system would produce nothing on a bad day. Instead the `GHS-BUILT-S R2023A` and OpenStreetMap path always works, so there is always a result.

5.5 Design decisions and reasons

Decision	Reason
Two acquisition paths	The core result exists before anyone logs in
Everything driven by a configuration file	A different city is a new file, not a code change — this is what makes the city-search goal reachable
Two matching grids	Keeps change detection sharp while keeping claims honest
Failed stage records itself and the run continues	A partial answer beats no answer
Learned norm rather than a fixed threshold	Transfers to a new city unchanged
Logistic Regression rather than deep learning	No graphics processing unit, no labels, readable coefficients, little accuracy given up
Colour palette validated for colour-vision deficiency	The intuitive green/red pairing measures ΔE 4.1 under deuteranopia — the two most important classes would be indistinguishable to a red-green colourblind reader

6. Proposed Solution

6.1 The end goal — city search to detailed analysis

1. User searches for a city
↓
2. System draws a circle around the city centre
(user-adjustable radius; automatic fitting later)
↓
3. System downloads satellite data for that circle
GHSL · VIIRS · Sentinel-2 · Landsat · OpenStreetMap
↓
4. Indicators computed for every grid cell
built-up · vegetation · temperature · night light · shops · roads
↓
5. Machine-learning models run on those indicators
↓
6. Dashboard presents
 - how the city has grown so far
 - where growth is predicted, +5 and +10 years
 - which development is underused

Why a circle. A city has no single official boundary available everywhere, and municipal limits routinely exclude the fastest-growing peri-urban belt — which is exactly where the findings concentrate. A circle around the centre is simple, works for any city, and can be widened by the user.

Current state. Steps 3–6 work today. Steps 1–2 are performed by editing `config/varanasi.yaml` rather than typing a name. Replacing that file with a search box changes only the centre point and radius; nothing downstream changes. This is the honest position and it is stated as such.

6.2 The complete pipeline — seven stages

Run with `python -m urbanintel.pipeline`.

#	Stage	Reads	Produces
1	<code>builtup</code>	<code>GHS-BUILT-S R2023A</code> , <code>GHS-POP R2023A</code>	Built-up per epoch, change, urban form, population
2	<code>osm</code>	OpenStreetMap	Point-of-interest density by sector, road density
3	<code>gee</code>	VIIRS, Sentinel-2, Landsat, Dynamic World, Open Buildings	Radiance and its trend, NDVI, NDBI, LST, land cover, building height
4	<code>vegetation</code>	NDVI from 3, built-up change from 1	Green lost, and how much became built-up
5	<code>thermal</code>	LST from 3	Heat-island intensity, hotspots, vulnerability
6	<code>ghost</code>	Radiance, points of interest, population, built-up	Activity index, expected-activity residual, six-class typology, zones
7	<code>export</code>	Everything above	500 m grid, 51 rasters, summary with provenance

The prediction model runs as a separate step, `scripts/run_growth_model.py`, because it consumes the pipeline's output as its input.

6.3 Models used, and why each

Model	Where used	Why this model
Binned median regression	Expected-activity curve (§3.6)	Non-parametric, so it adapts to any city's own relationship between building and activity without assuming a functional form. This is what replaces a fixed threshold
Ordinary least squares trend with significance test	Nighttime-light trend per cell	Twelve annual observations per cell; a linear slope with a p-value is sufficient and interpretable. Separates <code>emerging</code> from <code>ghost_growth</code>
Landscape expansion index	Urban form classification	A geometric rule, not a learned model — the classification is definitional, so learning it would add error without adding information
Logistic Regression (balanced class weights)	Growth suitability	Rare-event binary outcome; coefficients are directly readable, which matters for a planning audience
Constrained cellular automaton	Growth allocation	Enforces spatial contiguity that a per-cell model cannot express. Follows Liang et al. (2021)
Random-allocation baseline	Model benchmarking	Without it a Figure of Merit is uninterpretable — the reader cannot tell whether 0.07 reflects skill or the base rate

Models considered and not adopted

Model	Why not
U-Net or other convolutional network	Needs labelled training data and a graphics processing unit; literature reports parity with cellular automata for this task (§2.4). Recorded as a Phase 3 upgrade
Random forest for suitability	Would improve fit but has no view of a cell's surroundings and yields no readable coefficients. Planned as a Phase 3 comparison baseline, following Shojaei et al.
CA-Markov	Superseded by patch-based cellular automata in the current literature

6.4 How the datasets are approached

The approach is deliberately layered so that no single dataset failure destroys the result.

First, establish the physical facts. `GHS-BUILT-S R2023A` and `GHS-POP R2023A` give built-up surface and population with no account required. This alone yields expansion rates, urban form, and the land-versus-population ratio — the core finding of the project.

Second, add activity. VIIRS radiance, OpenStreetMap points of interest and population are combined into one index, each normalised per unit of built-up area. No single signal is trusted; agreement across three is what makes a low reading meaningful.

Third, add the trend. The twelve-year radiance series turns a static reading into a direction of travel, which is what separates stalled development from development filling up.

Fourth, add context. NDVI gives green-cover loss attributable to construction; LST gives heat-island intensity; Dynamic World cross-checks land cover and supplies the water mask; Open Buildings adds the vertical dimension.

Fifth, project forward. The seven drivers feed the suitability model, whose output the cellular automaton allocates.

Each tier is independently useful, and each degrades gracefully if its data is unavailable — recorded in `skipped` rather than silently dropped.

6.5 Dashboard

The delivered dashboard has six tabs:

Tab	Contents
Map	12 selectable layers over the reporting grid
Growth	Expansion by epoch, urban form breakdown, growth hotspots
Ghost growth	Typology areas, ranked zone table, zone locations
Environment	Green-cover loss, heat-island intensity and hotspots
Data	The full 4,765-cell grid, downloadable as CSV, plus the summary
Sources	Every dataset with its official identifier and actual imagery dates

A separate lightweight **dataset viewer** (`dataset_viewer/map.html`) presents each acquired dataset as a georeferenced overlay on a zoomable street map, with opacity control and per-layer acquisition dates. It requires no server and works offline.

7. Results Obtained So Far

7.1 Urban expansion, 2010–2020 (observational)

Year	Built-up surface	Urban extent (≥20% built)	Population
2010	72.48 km ²	128.43 km ²	4.17 M
2015	81.05 km ²	142.25 km ²	4.41 M
2020	89.73 km ²	154.39 km ²	4.60 M

- Built-up surface **+23.8%** over ten years
- Urban extent **+1.40% per year** compound
- Population **+10.3%**
- **Land consumed 2.3× faster than population** — the single most policy-relevant number in this project

7.2 Form of new development

Of 13.45 km² that newly crossed the urban threshold:

Form	Area	Share
Infill	0.95 km ²	7.1%
Edge expansion	5.94 km ²	44.2%
Leapfrog	6.56 km²	48.8%

Nearly half of new development is detached from the existing urban fabric.

7.3 Growth typology (500 m reporting grid)

Class	Area
Undeveloped	1,101.09 km ²
Established active	140.69 km ²
Healthy growth	4.54 km ²
Emerging	6.88 km ²
Declining	5.70 km ²
Ghost growth	0.35 km² across 15 zones

7.4 The result that validates the method

Before the nighttime-light series was integrated, the system flagged **7.24 km²** and labelled it an upper bound, stating in advance that dim-but- brightening cells could not be separated from dim-and-static ones. Adding the 2013–2024 series split it almost exactly:

old ghost_growth	7.24 km ²	
└─ ghost_growth	0.35 km ²	dim and NOT rising
└─ emerging	6.88 km ²	dim but BRIGHTENING – filling up

	7.23 km ²	

95% of what looked like stalled development is a neighbourhood mid-occupation. A new class also appeared that could not exist before: **5.70 km²** declining, drawn entirely out of established_active (146.39 → 140.69 km², an exact match).

All 15 remaining zones are peripheral, consistent with 48.8% leapfrog growth — two independent analyses agreeing.

7.5 Activity, green cover and heat

Metric	Value
Sum of lights, 2013 → 2024	518,008 → 905,354 (+ 74.8%)
Lit fraction of the area, 2024	95.5%
Green cover, 2018 → 2024	71.0% → 77.9%
Green lost	6.37 km ²
Green lost specifically to built-up conversion	0.05 km²
Rural reference temperature, pre-monsoon 2024	41.12 °C
Mean urban heat-island intensity	+1.10 °C
Maximum intensity	+9.27 °C
Hotspot area (≥ +3 °C)	26.57 km ²

The gross green-cover figure is not reported as a finding. The October–March window is the *rabi* cropping season, so NDVI across an agricultural area tracks the wheat and pulse crop far more than urban canopy. Only lost_to_builtup is robust, because intersecting NDVI loss with built-up gain removes anything that is merely a cropping-calendar difference.

7.6 Predictive model validation

Metric	Value
AUC (Area Under the Curve), training	0.9901
Figure of Merit , held-out 2015→2020	0.0679
Random-allocation baseline	0.0055
Skill versus random	12.3×
Cohen's κ	0.1176
Hits / observed conversions	154 / 1,214
Projected urban extent, 2030	+31.21 km²

All seven coefficient signs are theoretically correct — distance from centre -0.1410 , distance from urban edge -0.1438 , built-up fraction $+2.2616$.

Honest reading. A Figure of Merit of 0.068 means roughly one in fifteen predicted conversions is correct, below the 0.1–0.3 typical of published land-change models and well below the 0.2642 reported by Liang et al. The model carries real signal — 12.3× better than chance — but is **not accurate enough for parcel-level use**, and is not presented as such.

An independent check worth reporting. Our 2030 projection was compared against GHSL's own 2025 projection — same 2020 starting state, comparable demand. They agree at **Figure of Merit 0.0013**, essentially not at all. Two models starting from identical conditions place growth in almost entirely different locations. This independently justifies excluding the projected epochs, and is a real caution about treating modelled land-cover products as data.

7.7 Verification

- **25 of 25 automated tests pass**, covering grid nesting, area conservation under aggregation, the priority reducer, urban form classification, trend recovery, the ghost screen on synthetic data with a known answer, and the land-change metric accounting
 - **Pipeline runs end to end with no skipped layers**, producing 51 raster layers, a 4,765-cell reporting grid, and a summary carrying full provenance
 - **Dashboard renders with 0 exceptions**, 6 tabs
 - **Colour palette validated** with a colour-vision-deficiency checker rather than chosen by eye
-

8. Technical Challenges and Corrective Action

Seven defects were found and fixed. Four were caught by disbelieving an implausible result rather than by a test — which is why they are recorded here rather than quietly corrected.

#	Defect	How it would have corrupted the output
1	Frame misalignment. The 100 m and 500 m frames were built by independent calls, each snapping outward to its own resolution	They did not tile each other; aggregation crashed on a shape mismatch. Had the dimensions happened to agree, every aggregate would have been silently misaligned instead. Fixed with <code>AOI.frame_pair()</code>
2	Resolution-dependent threshold. <code>min_builtup_m2: 5000</code> was documented as "2% of a 500 m cell" but applied on the 100 m frame, where it means 50%	The activity index was defined for only 1,418 of 15,816 urban cells. Now expressed as a fraction of cell area
3	Majority aggregation erased the findings. Ghost cells are a minority by nature	A 500 m majority vote wiped every one out of the reporting grid. Replaced with a <code>priority</code> reducer that surfaces the most significant class present; regression test added
4	Perfect linear fit mapped to $p = 1$. Zero residual gave $t = 0$	Discarded exactly the cleanest nightlight trends. Fixed to treat a perfect fit as maximally significant
5	GHSL 2025 and 2030 treated as observations. They are the GHSL model's own projections	The model was being validated against another model's forecast, and Phase 1 headline figures presented a forecast as measurement. All fitting now restricted to 1975–2020
6	Nine of twelve nightlight years silently missing. The whole series pointed at <code>ANNUAL_V22</code> , which holds only 2022 onward	Every pre-2022 year resolved to an <i>empty</i> collection. The trend separating <code>ghost_growth</code> from <code>emerging</code> would have been fitted on three years instead of twelve. Fixed with asset-resolution by year, plus an explicit error when a year yields zero images
7	Vegetation baseline was one day of imagery. The Oct 2015 – Mar 2016 window holds 3 scenes, all from 2015-12-28	Differencing it against a 148-scene composite produced an apparent green-cover rise from 27.9% to 77.9% — entirely an artefact of sampling depth. Baseline moved to 2018 and a minimum composite-depth check added

Defects 6 and 7 share a shape worth naming. Neither threw an exception, both produced numbers of the right order of magnitude, and the vegetation one would have been reported as a striking positive finding. They were caught by checking provenance against expectation — asking whether a 50-point rise in green cover in six years was physically plausible — not by any test. Both now have **structural guards** so the class of error cannot recur silently for any index, year or city.

9. Limitations and Future Work

9.1 What this system is not

- **A screening tool, not a census.** VIIRS resolves about 460 m and cannot see one empty housing block. The unit of a reliable finding is a neighbourhood.
- **Unvalidated against ground truth.** Nothing here has been checked against observation. At 0.35 km², some zones are four 50 m cells — small enough that a single mapping error could produce one.
- **Dependent on an uneven OpenStreetMap.** Varanasi's core is well mapped; the periphery is not, and all 15 zones are peripheral.
- **Built on a modelled product.** `GHS-BUILT-S R2023A` is modelled, not measured, and is documented to under-detect low-rise informal development.
- **Carrying a version discontinuity** at 2021/2022 in the nightlight series.
- **Reporting a modest predictive score.** Figure of Merit 0.0679 is real signal but not parcel-accurate.

Zones 14 and 15, at scores 0.56 and 0.50 with the shallowest residuals, are the boundary of the method's discrimination and should be presented as such rather than as findings of equal standing.

9.2 Future work

Priority	Task	Why
1	Ground validation of the 15 zones	Converts a screening output into an evidenced finding; the single highest-value remaining step
2	Ward-level reporting on Census 2011 Varanasi Nagar Nigam boundaries (90 wards)	Municipal planners act on wards, not a 500 m grid
3	City search box replacing the configuration file	Delivers the stated end goal; nothing downstream changes
4	Building height from Open Buildings Temporal	Distinguishes densification from outward spread, and a tall unlit building is a far stronger signal than area alone
5	Random-forest baseline for the suitability model	Like-for-like comparison following Shojaei et al.
6	Master-plan overlay from the Varanasi Development Authority	Yields development where none was planned, and planned development that never materialised — the sharpest available framing
7	Multi-city comparison	The configuration is already city-agnostic

10. References

1. Chen, X., Wang, Z., Zhang, F., Shen, G. & Chen, Q. (2024). A global annual simulated VIIRS nighttime light dataset from 1992 to 2023. *Scientific Data* **11**, 1380. DOI 10.1038/s41597-024-04228-6
2. Tian, Y., Cheng, K. M., Zhang, Z. et al. (2026). An Extended VIIRS-like Artificial Nighttime Light Data Reconstruction (1986–2024). *Scientific Data* **13**, 233. DOI 10.1038/s41597-026-06549-0
3. Chen, Z. et al. (2021). An extended time series (2000–2018) of global NPP-VIIRS-like nighttime light data from a cross-sensor calibration. *Earth System Science Data* **13**, 889–906. DOI 10.5194/essd-13-889-2021
4. Li, X., Zhou, Y., Zhao, M. & Zhao, X. (2020). A harmonized global nighttime light dataset 1992–2018. *Scientific Data* **7**, 168. DOI 10.1038/s41597-020-0510-y
5. Zhang, L., Ren, Z., Chen, B., Gong, P., Xu, B. & Fu, H. (2024). A Prolonged Artificial Nighttime-light Dataset of China (1984–2020). *Scientific Data* **11**, 414. DOI 10.1038/s41597-024-03223-1
6. Nechaev, D., Zhizhin, M., Poyda, A., Ghosh, T., Hsu, F.-C. & Elvidge, C. (2021). Cross-Sensor Nighttime Lights Image Calibration for DMSP/OLS and SNPP/VIIRS with Residual U-Net. *Remote Sensing* **13**(24), 5026. DOI 10.3390/rs13245026
7. Goldblatt, R., Stuhlmacher, M. F., Tellman, B. et al. (2018). Using Landsat and nighttime lights for supervised pixel-based image classification of urban land cover. *Remote Sensing of Environment* **205**, 253–275.
8. Marconcini, M., Metz-Marconcini, A., Üreyen, S. et al. (2020). Outlining where humans live, the World Settlement Footprint 2015. *Scientific Data* **7**, 242. DOI 10.1038/s41597-020-00580-5
9. Tang, Y., Shao, Z., Huang, X. & Cai, B. (2021). Mapping Impervious Surface Areas Using Time-Series Nighttime Light and MODIS Imagery. *Remote Sensing* **13**(10), 1900. DOI 10.3390/rs13101900
10. Sirko, W., Kashubin, S., Ritter, M. et al. (2021). Continental-Scale Building Detection from High Resolution Satellite Imagery. *arXiv:2107.12283*
11. Brown, C. F., Brumby, S. P., Guzder-Williams, B. et al. (2022). Dynamic World, Near real-time global 10 m land use land cover mapping. *Scientific Data* **9**, 251. DOI 10.1038/s41597-022-01307-4
12. Zhang, Y., Tu, T. & Long, Y. (2024). Inferring ghost cities on the globe in newly developed urban areas based on urban vitality with multi-source data. *arXiv:2408.15117*; published in *Cities* (2025).
13. Yeh, C., Perez, A., Driscoll, A. et al. (2020). Using publicly available satellite imagery and deep learning to understand economic well-being in Africa. *Nature Communications* **11**, 2583. DOI 10.1038/s41467-020-16185-w
14. Anucharn, T., Hongpradit, P., Iamchuen, N. & Puttinaovararat, S. (2025). Spatial Analysis of Urban Expansion and Energy Consumption Using Nighttime Light Data. *ISPRS International Journal of Geo-Information* **14**(4), 178. DOI 10.3390/ijgi14040178
15. Liang, X., Guan, Q., Clarke, K. C., Liu, S., Wang, B. & Yao, Y. (2021). Understanding the drivers of sustainable land expansion using a patch-generating land use simulation (PLUS) model. *Computers, Environment and Urban Systems* **85**, 101569. DOI 10.1016/j.compenvurbsys.2020.101569
16. Chen, G., Li, X., Liu, X. et al. (2020). Global projections of future urban land expansion under shared socioeconomic pathways. *Nature Communications* **11**, 537. DOI 10.1038/s41467-020-14386-x
17. Shojaei, H., Nadi, S., Shafizadeh-Moghadam, H., Tayyebi, A. & Van Genderen, J. (2022). An efficient built-up land expansion model using a modified U-Net. *International Journal of Digital Earth* **15**(1), 148–163.

DOI 10.1080/17538947.2021.2017035

18. Zhang, A., Tariq, A., Quddoos, A. et al. (2025). Spatio-temporal analysis of urban expansion and land use dynamics using google earth engine and predictive models. *Scientific Reports* **15**, 6993. DOI 10.1038/s41598-025-92034-4
19. Ermida, S. L., Soares, P., Mantas, V., Götsche, F.-M. & Trigo, I. F. (2020). Google Earth Engine Open-Source Code for Land Surface Temperature Estimation from the Landsat Series. *Remote Sensing* **12**(9), 1471. DOI 10.3390/rs12091471
20. Ramachandra, T. V., Rana, R. S., Vinay, S. & Aithal, B. H. (2025). Urban heat island linkages with the landscape morphology. *Scientific Reports* **15**, 24485. DOI 10.1038/s41598-025-09141-5

Cited in support

21. Liu, X., Li, X., Chen, Y., Tan, Z., Li, S. & Ai, B. (2010). A new landscape index for quantifying urban expansion using multi-temporal remotely sensed data. *Landscape Ecology* **25**, 671–682.
22. Pontius, R. G. et al. (2008). Comparing the input, output, and validation maps for several models of land change. *Annals of Regional Science* **42**, 11–37. DOI 10.1007/s00168-007-0138-2
23. Henderson, J. V., Storeygard, A. & Weil, D. N. (2012). Measuring Economic Growth from Outer Space. *American Economic Review* **102**(2), 994–1028.

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